CreditMetrics Methodology and Credit Value at Risk (Credit VaR)
Author: Dr. Yogesh Malhotra

Described by Hull (2011, 2012) as 'a procedure for calculating credit value at risk', **CreditMetrics** methodology (RiskMetrics Group 2007) is used for assessing portfolio risk due to changes in bond or debt value caused by credit quality changes including *credit migration* (upgrades and downgrades), as well as, default. It measures the uncertainty in forward value of the bond portfolio at the risk horizon caused by such *credit events*. Changes in debt value could be small in case of credit quality ratings change; however, they could be enormous, 50% to 90%, in case of default. Characterized by a long downside tail, credit-returns are highly-skewed and fat-tailed and thus far from the Gaussian *normal distribution* assumptions about market risk in VaR (Fig. 1). In the portfolio context, based upon correlation of credit quality moves across obligors, CreditMetrics assesses both value-at-risk (VaR), i.e., the volatility of value, as well as expected losses (EL). By distinguishing high quality well-diversified portfolios from low-quality concentrated portfolios, it offers better understanding of credit risk in terms of *diversification benefits* and *concentration risk* compared to mandated standard capital adequacy measures.

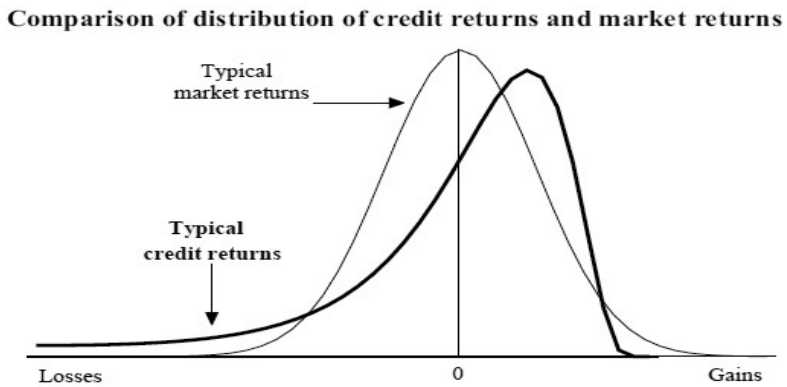

Figure 1. *Source*: **CreditMetrics Technical Document**, RiskMetrics Group (2007).

Framework of CreditMetrics Methodology

The framework of the CreditMetrics methodology discussed in the CreditMetrics Technical Document (Riskmetrics Group, 2007) and reviewed at different levels in Crouhy (2001) and Hull (2011, 2012) is shown on the next page. We will apply this methodology to illustrate computation of 1 year, 99% credit value at risk for a 3-bond portfolio comprised of AA, A, and B rated bonds. In these illustrations focused on key steps in the methodology rather than specific computations, 40% loss, i.e., 60% recovery is assumed in case of default of any bond. The key steps in the CreditMetrics methodology consistent with Riskmetrics Group (2007), Crouhy (2001), and, Hull (2011, 2012) are listed below after the schematic description of the CreditMetrics framework and discussed subsequently.

Electronic copy available at: https://ssrn.com/abstract=3862641

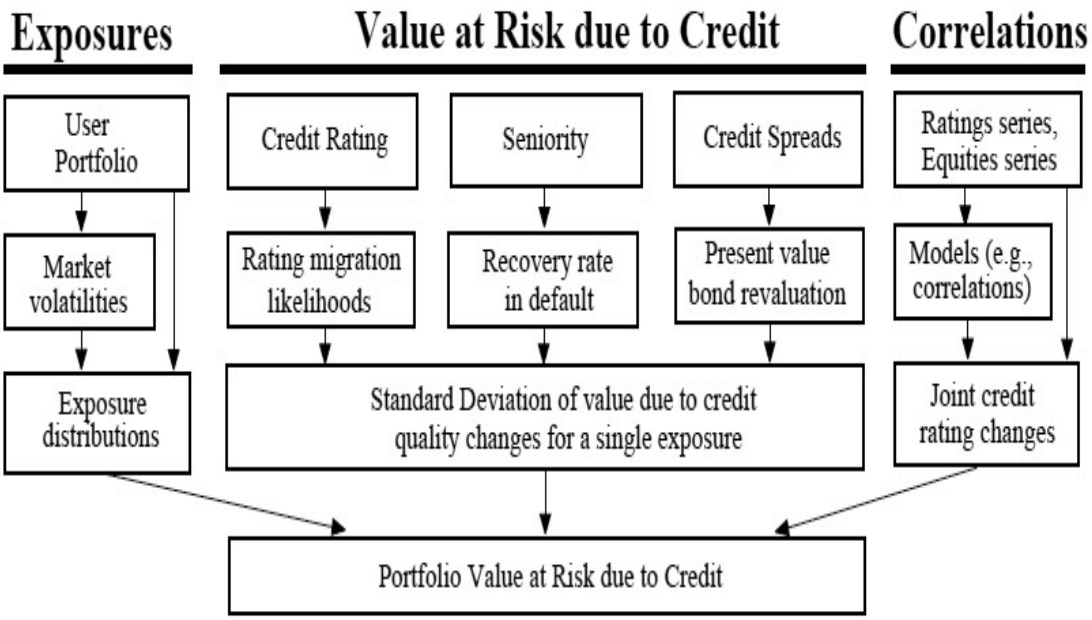

Figure 2. *Source*: **CreditMetrics Technical Document**, RiskMetrics Group (2007).

Key steps in the CreditMetrics methodology focused Credit Value at Risk (Credit VaR) are:

(A) Credit VaR Computation for each Bond
  Step 1: Specification of the Credit Risk Horizon
  Step 2: Specification of the Transition Matrix
  Step 3: Specification of the Forward Pricing Model
  Step 4: Derivation of Forward Distribution of Changes in Portfolio Value
(B) Portfolio Credit VaR Computation of All Bonds
(C) Correlations Analysis of Diversification Benefits Using Simulation
(D) Credit VaR and Calculation of Economic Capital
(E) Marginal Risk Contribution Analysis of Diversification Benefits

(A) Credit VaR Computation for each Bond:

Step 1: Specification of the Credit Risk Horizon: Given our objective of computing 1 year, 99% Credit VaR, we will use a credit risk horizon of one year. In other words, we want to list all possible credit outcomes that a specific bond will meet at the end of one year given specific credit events.

Step 2: Specification of the Transition Matrix: Given any specific time horizon for determining credit risk, the specific bond or credit (issuer or name) will meet one of the following fates: (i) its credit quality rating will be same at the end of the period; (ii) its credit quality rating will either be upgraded or downgraded; (iii) it will default. Each of such outcomes for any specific bond has a different probability of occurrence which can be derived based upon historical data. The specific outcomes are listed in a table called *transition matrix* that shows initial rating at beginning of the

Electronic copy available at: https://ssrn.com/abstract=3862641

year in left column and rating at yearend in the top row. The transition matrix should be estimated for the same time interval as the risk horizon that is established in above Step 1.

One-year transition matrix (%)

Initial rating	Rating at year-end (%)							
	AAA	AA	A	BBB	BB	B	CCC	Default
AAA	90.81	8.33	0.68	0.06	0.12	0	0	0
AA	0.70	90.65	7.79	0.64	0.06	0.14	0.02	0
A	0.09	2.27	91.05	5.52	0.74	0.26	0.01	0.06
BBB	0.02	0.33	5.95	86.93	5.30	1.17	0.12	0.18
BB	0.03	0.14	0.67	7.73	80.53	8.84	1.00	1.06
B	0	0.11	0.24	0.43	6.48	83.46	4.07	5.20
CCC	0.22	0	0.22	1.30	2.38	11.24	64.86	19.79

Source: Standard & Poor's CreditWeek (15 April 96)

Table 1. *Source*: **CreditMetrics Technical Document**, RiskMetrics Group (2007).

The intersection of the specific row and column item can then be used to determine what will be the end-of-year rating for any specific initial rating for a given bond. Using the transition matrix from the CreditMetrics Technical Document, we illustrate this computation based upon the given bond portfolio data for 3-bond portfolio of AA, A, and B rated bonds.

The most probable outcome at the yearend for any bond is that it will remain at the same credit quality rating as at the beginning, hence, the specific probabilities of such outcomes for the three bonds are listed below:

AA	90.65%
A	91.05%
B	83.46%

As apparent, a bond of lower quality credit rating has a greater probability of default relative to others. From above data, we see that B-rated bond has more than 5% probability of default in contrast whereas A-rated has 0.06% and AA-rated has near 0% probability respectively.

Similarly, AA has 0.70% probability of upgrade and 7.79% of downgrade; A has 2.27% probability of upgrade and 5.52% of downgrade; and B has 6.48% of upgrade and 4.07% of downgrade. For any given bond-rating, there are in all eight outcomes with different probabilities all of which sum together to 100%.
Assuming that the specific bonds' credit quality ratings are statistically independent, i.e., uncorrelated, the combined or joint probability of the three outcomes will be given by their

Electronic copy available at: https://ssrn.com/abstract=3862641

product. For example, in the above case of the three bond ratings remaining same at the yearend is given by their product 90.65% x 91.05% x 83.46%, or, 68.89%. Given that each of the three bonds has 8 possible yearend outcomes, the total number of yearend outcomes is given by 8x8x8, or, 512.

Step 3: Specification of the Forward Pricing Model: The value of any bond at the end of the risk horizon, one year in our case, is the present value of the remaining cash flows. Such cash flows are made up of coupon payments and payment of principal at maturity. The value of the bond is computed by discounting those cash flows using discount rates derived from the forward zero curve for each specific rating category. These forward curves are computed as of the end of the risk horizon and for each specific rating category determine the value of the bond upon upgrade or downgrade. Using the one-year forward zero curve data from the CreditMetrics Technical Document, we illustrate this computation based upon the given bond portfolio data for 3-bond portfolio of AA, A, and B rated bonds.

Example one-year forward zero curves by credit rating category (%)

Category	Year 1	Year 2	Year 3	Year 4
AAA	3.60	4.17	4.73	5.12
AA	3.65	4.22	4.78	5.17
A	3.72	4.32	4.93	5.32
BBB	4.10	4.67	5.25	5.63
BB	5.55	6.02	6.78	7.27
B	6.05	7.02	8.03	8.52
CCC	15.05	15.02	14.03	13.52

Table 2. *Source*: **CreditMetrics Technical Document**, RiskMetrics Group (2007).

To illustrate how the forward curve pricing model is used to compute one-year forward value of A bond, we compute one-year forward bond price for a (say) 3-year A bond with (say) annual coupon of 5% that remains A-rated at yearend:

Forward Value: $5 + 5/1.0372 + 105/(1.0432)^2 = 106.30$

Using similar computations for all possible eight year-end ratings for the A-bond, we can find one-year forward value of A bond for all possible 8 rating outcomes (including default).

Electronic copy available at: https://ssrn.com/abstract=3862641

Year-end values after credit rating migration from single-A ($)

Year-end rating	Coupon	Forward Value	Total Value
AAA	5.00	101.59	106.59
AA	5.00	101.49	106.49
A	5.00	101.30	106.30
BBB	5.00	100.64	105.64
BB	5.00	98.15	103.15
B	5.00	96.39	101.39
CCC	5.00	73.71	88.71
Default	–	51.13	51.13

Table 3. *Source*: **CreditMetrics Technical Document**, RiskMetrics Group (2007) (Adapted).

By computing the credit rating migration from second and third bond, similar year-end values after credit rating migration are determined. For a two-bond portfolio, by creating a table with year-end values of bond one on top row, and bond two year-end values in left column, a matrix of sums at intersections of rows and column values yields all possible 8x8 (64) year-end values. Likewise, for a 3-bond portfolio, a 3-D matrix with year-end values for three dimensions is created yielding 8x8x8 (512) possible year-end values for the complete portfolio.

Step 4: Derivation of Forward Distribution of Changes in Portfolio Value: The eight possible probable *credit event* outcomes for A bond shown in Table 1 are listed below.

Probability of credit rating migrations in one year for a single-A

Year-end rating	Probability (%)
AAA	0.09
AA	2.27
A	91.05
BBB	5.52
BB	0.74
B	0.60
CCC	0.01
Default	0.06

Table 4. *Source*: **CreditMetrics Technical Document**, RiskMetrics Group (2007).

By multiplying corresponding values from Table 3 (year-end values after credit rating migrations from Single-A) and Table 4 (probability of credit rating migration in one year for a single-A), we can obtain the probability weighted forward distribution of changes in portfolio value.

Electronic copy available at: https://ssrn.com/abstract=3862641

(B) Portfolio Credit VaR Computation of All Bonds:

Assuming that the specific bonds' credit quality ratings are statistically independent, i.e., uncorrelated, the combined or joint probability of the three outcomes will be given by their product. As noted before, in the above case probability of the three bond ratings remaining same at the yearend is given by their product 90.65% x 91.05% x 83.46%, or, 68.89%. Given that each of the three bonds has 8 possible yearend outcomes, the total number of yearend outcomes is given by 8x8x8, or, 512.

In reality, however, the changes in the three bond ratings quality are not independent, i.e., the default correlations between them are not zero. Credit VaR is very sensitive to default correlations which may increase during a recession and decrease in good economic times. CreditMetrics uses Gaussian copula correlation between equity returns as a proxy for asset returns correlations and their joint distributions to infer correlations between changes in credit quality (Hull 2011). The framework for thinking about default as a function of underlying volatile asset value of the firm was first proposed by Merton (1974) and known as option theoretic valuation of debt. Building upon the Black and Scholes option pricing model it values credit risk component of a firm's debt like a put option on the value of the underlying. Default threshold is the point at which the value of firm's assets is less than its debts outstanding.

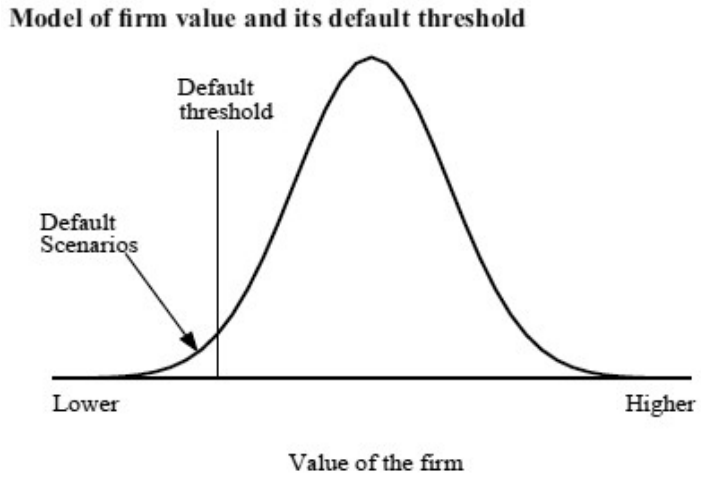

Figure 3. *Source*: **CreditMetrics Technical Document**, RiskMetrics Group (2007).

Extending the Merton model to include ratings changes involves in addition to the default threshold identifying credit rating upgrade and downgrade thresholds which determine future rating as determined by firm's asset value.

Calculating a one-year credit VaR for a bond portfolio involves carrying out a Monte Carlo simulation of ratings transitions over the same duration. Based on simulation trials the end of year credit ratings are determined and bonds revalued. Determination of total credit losses for the year is then used to find the 99% worst result characterized as the one-year 99% credit VaR for the portfolio.

Electronic copy available at: https://ssrn.com/abstract=3862641

(C) Correlations Analysis of Diversification Benefits Using Simulation:

The simulation of our A-rated and B-rated bond over a one-year period using the following transition matrix illustrates how to factor in default correlations represented in terms of copula correlation between equity returns of (say) 0.2 (Hull 2011).

Table 15.1 One-year ratings transition matrix, 1970–2007, with probabilities expressed as percentages and adjustments for transition to the without-rating (WR) category. *Source*: Moody's.

Initial rating	*Rating at year end*								
	Aaa	Aa	A	Baa	Ba	B	Caa	Ca–C	Default
Aaa	91.37	7.59	0.85	0.17	0.02	0.00	0.00	0.00	0.00
Aa	1.29	90.84	6.85	0.73	0.19	0.04	0.00	0.00	0.07
A	0.09	3.10	90.23	5.62	0.74	0.11	0.02	0.01	0.08
Baa	0.05	0.34	4.94	87.79	5.54	0.84	0.17	0.02	0.32
Ba	0.01	0.09	0.54	6.62	82.76	7.80	0.63	0.06	1.49
B	0.01	0.06	0.20	0.73	7.10	81.24	5.64	0.57	4.45
Caa	0.00	0.03	0.04	0.24	1.04	9.59	71.50	3.97	13.58
Ca–C	0.00	0.00	0.14	0.00	0.55	3.76	8.41	64.19	22.96
Default	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	100.00

Table 5. *Source*: **Moody's**, Hull (2011).

On each simulation trial, two variables x_A and x_B with correlation of 0.2, representing new ratings of A- and B-rated bonds respectively, are sampled from standard normal distributions. The corresponding rating upgrade and downgrade thresholds can then be determined by finding the inverse of normal probability distribution function for respective values in the transition matrix. For example, for initial rating of A, the numbers of normal standard deviations corresponding to first three yearend ratings are given by respective inverses of normal pdf:

$N^{-1}(0.09\%) = N^{-1}(0.0009) = -3.1214$

⇨ A-bond gets upgraded to Aaa if $x_A < -3.1214$;

$N^{-1}(0.09\% + 3.10\%) = N^{-1}(0.0009 + 0.0310) = -1.8536$

⇨ A-bond gets upgraded to Aa if $-3.1214 < x_A < -1.8536$;

$N^{-1}(0.09\% + 3.10\% + 90.23\%) = N^{-1}(0.0009 + 0.0310 + 0.9023) = 1.5078$

⇨ A-bond stays rated A if $-1.8536 < x_A < 1.5078$.

A-rated bond defaults if $x_A < N^{-1}(0.0008)$, i.e., $x_A > N^{-1}(0.9992)$, i.e., $x_A > 3.1559$.

Similarly,

$N^{-1}(0.01\%) = N^{-1}(0.0001) = -3.7190$

⇨ B-bond gets upgraded to Aaa if $x_B < -3.7190$;

$N^{-1}(0.01\% + 0.06\%) = N^{-1}(0.0001 + 0.0006) = -3.1947$

⇨ B-bond gets upgraded to Aa if $-3.7190 < x_B < -3.1947$;

Electronic copy available at: https://ssrn.com/abstract=3862641

$N^{-1}(0.01\% + 0.06\% + 0.20\%) = N^{-1}(0.\ 0001 + 0.\ 0006 + 0.0020) = -2.7822$

⇨ B-bond gets upgraded to A if $-3.1947 < x_A < -2.7822$.

B-rated bond defaults if $x_B < N^{-1}(0.0445)$, i.e., $x_B > N^{-1}(0.955)$, i.e., $x_B > 1.7007$.

The effect of asset return correlation on the joint default probability for our two bonds is shown in Figure 4 below.

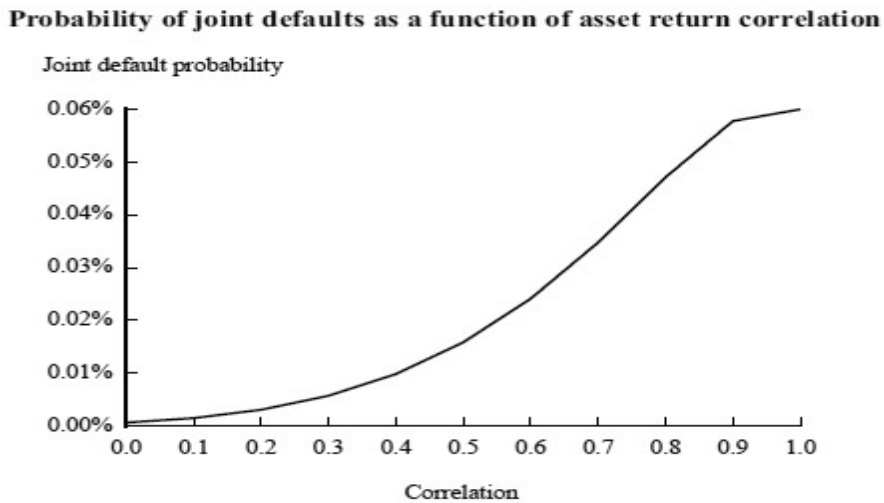

Figure 4. *Source*: **CreditMetrics Technical Document**, RiskMetrics Group (2007).

On each simulation trial, besides rating change of each bond, idiosyncratic spread change is examined by using a one-factor regression model to divide the spread component of each bond into systematic and idiosyncratic components. If credit rating remains same, bond revaluation reflects the idiosyncratic spread change for the bond. If credit rating changes, bond revaluation uses spread corresponding to the new category. If default occurs, a recovery rate is applied.

(D) Credit VaR and Calculation of Economic Capital:

The risk-taking capability or the financial cushion that a bank uses to absorb unexpected losses such as those related to credit events is called *economic capital*.

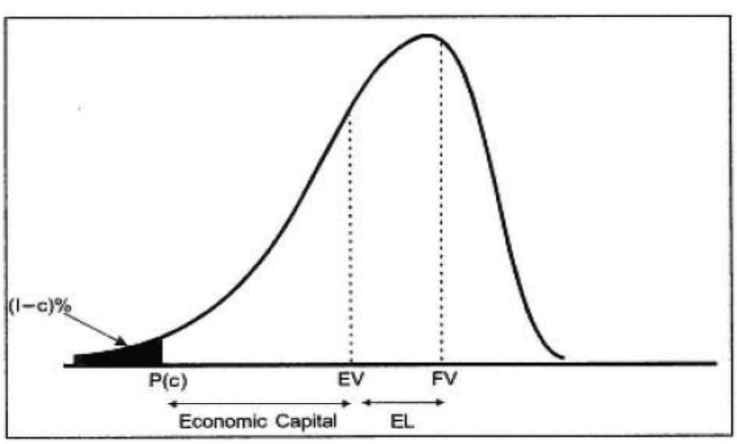

Figure 5. Credit VaR and Calculation of Economic Capital *Source*: Crouhy et al. (2001).
Capital = EV – P(c)
EV = Expected Value of Portfolio,
P(c) = Value of Portfolio in worst case at (1 – c%) confidence level,
FV = Forward Value of Portfolio,
EL = Expected Loss = FV – EV,
PR = Promised return on capital,
ER = Expected return on portfolio,
V_0 = Current mark-to-market value of portfolio,
$FV = V_0(1 + PR)$, $EV = V_0(1 + ER)$

Electronic copy available at: https://ssrn.com/abstract=3862641

Funds equivalent to EL are held in bank's reserves based upon the computations in Figure 5 showing how capital charge related to credit risk can be derived. Reserves need to be held against these unexpected losses at the given confidence level of 99%, so that we are 99% certain that the loss level during the given time horizon of one year will not exceed such losses.

(E) Marginal Risk Contribution Analysis of Diversification Benefits:

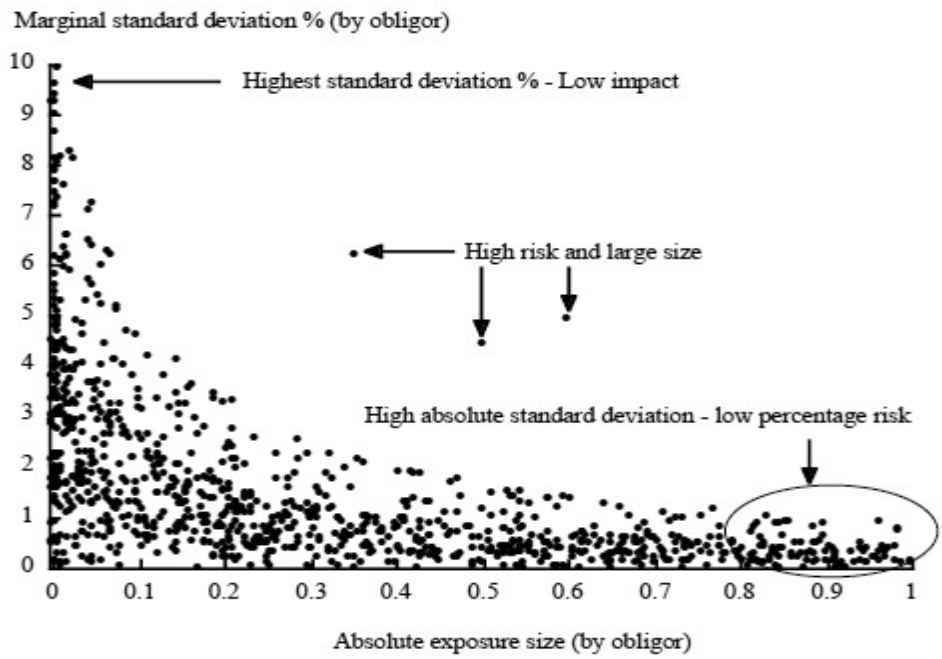

Figure 6. *Source*: **CreditMetrics Technical Document**, RiskMetrics Group (2007).

Marginal risk, which grows geometrically with concentration on specific debts, needs to be identified in order to identify over-concentrations and potential risk-mitigation actions. In other words, an exposure's risk contribution to a portfolio needs to be measured, accounting for diversification benefits as well as concentration risks. Marginal standard deviation of any specific debt or bond is the difference between the entire portfolio's standard deviation and the standard deviation of the same portfolio by excluding the specific debt or bond. Based on mapping the specific portfolios as in Figure 6, three broad categories of risks can be distinguished indicated by the three clusters with arrows. Of the three categories – largest absolute exposure size, highest percentage level of risk, and largest absolute amount of risk – the last category needs particular attention given its greatest contribution to portfolio volatility.

References:

1. Crouhy, M., Galai, D., and, Mark, R. Risk Management, McGraw-Hill, 2001.
2. Hull, J. Options, Futures, and Other Derivatives, 8th Edition, Prentice Hall, 2012.
3. Hull, J. Risk Management and Financial Institutions, 2nd Edition, Pearson, 2011.
4. Merton, Robert C. "On the Pricing of Corporate Debt: The Risk Structure of Interest Rates," *The Journal of Finance,* Vol. 29, (1974), pp. 449-470.
5. RiskMetrics Group. CreditMetrics Technical Document, RiskMetrics Group, 2007. https://www.msci.com/documents/10199/93396227-d449-4229-9143-24a94dab122f

Electronic copy available at: https://ssrn.com/abstract=3862641